

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
 April - May 2016

Duration: 3 Hrs
Semester: III

Max. Marks: 100

Class: SE

Name of the Course: Discrete Structures and Graph Theory

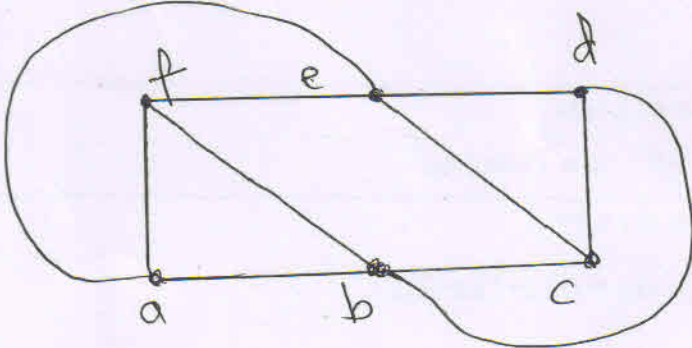
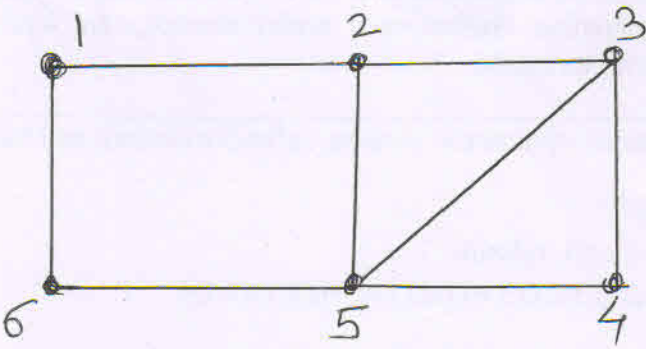
Branch: Comp

Course Code: UCEC305

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 a)	Among the integers 1 and 300 1) How many are divisible by 3,5 or 7 and how many are not divisible by 3,5 or 7? Justify with Venn diagram.	5
Q 1 b)	Show that using Laws of Logic [(P → Q) ∧ ~Q] → ~P is a Tautology	5
Q 1 c)	Prove by induction that for $n \geq 1$. $1.2 + 2.3 + \dots + n(n+1) = (n(n+1)(n+2))/3$.	5
Q 1 d)	State pigeon hole principle . Explain with example.	5
Q2 a)	Let $A = \mathbb{Z}$, and let R be defined as $aRb \mid a-b$ is an odd positive integer. Is R symmetric, transitive, reflexive , antisymmetric, an equivalence relation. Justify with example.	10
Q2 b)	Explain with example symmetric closure, reflexive closure and transitive closure. Let $A = \{1,2,3,4,5\}$ and relation $R = \{(1,2), (1,4), (2,2), (3,2), (3,4), (4,1), (4,3), (5,1), (5,2)\}$ Find the transitive closure of R by Warshall's Algorithm.	10
	OR	
Q2 a)	Let $A = \{a,b,c,d,e\}$ and $R = \{(a,a), (a,b), (a,c), (b,c), (c,e), (c,d), (d,e), (a,d)\}$ Compute R^2 , MR^2 , MR^3 draw digraph. Find connectivity Relation	10

	Use diagram method and justify the answer..	
Q2 b)	Let $A = (1, 2, 3, 4, 5)$, R is a relation defined as aRb iff $b \leq a+1$. Determine whether R is reflexive, irreflexive, symmetric, asymmetric, antisymmetric? Justify with example.	10
Q3 a)	Draw the Hasse diagram for D_{42} . Is it linear Lattice? Show all the steps of construction.	10
Q3 b)	Construct the Hasse diagram of D_{30} . Is it distributive Lattice?	10
	OR	
Q3 a)	What are the different conditions for complemented Lattice? Find the complement of each element in D_{30} .	10
Q3 b)	What is Boolean Algebra. Explain with example.	10
Q4 (a)	Find out Hamiltonian path and cycle. Justify.	5
		
Q4 (b)	Find out Euler path and circuit for the following graph Justify.	5
		
Q4 c)	What are the different conditions for the graph to be isomorphic? Explain with example.	5
Q4 d)	What are the conditions for planer graph. Justify with example	5

Q 5 a)	Let $H = \begin{matrix} 011 \\ 011 \\ 100 \\ 010 \\ 001 \end{matrix}$ Be parity matrix. Determine the $(2,5)$ group code function $eH : B^2 \rightarrow B^5$	10
Q 5 b)	Prove that the set $G = \{ 1,2,3,4,5,6 \}$ is a finite abelian group of order 6 with respect to multiplication modulo 7. Is it cyclic group ?	10